8. NMAP TO DISCOVER LIVE HOSTS USING NMAP SCANS

Ex. No: 8
Date: 4-9-26

Aim:

To discover live hosts on a network using Nmap host-discovery scans - ARP scan, ICMP scan, and TCP/UDP ping scan - and to understand which protocol operates at which layer of the TCP/IP model, and when each one applies.

N-map:

Nmap ("Network Mapper") is a free, open-source network scanning tool created by Gordon Lyon and first released in 1997. It is the industry-standard tool for mapping networks, discovering live hosts, and identifying the services running on them.

Procedure:

1. Log into TryHackMe and open the "Nmap Live Host Discovery" room.
2. Work through tasks 1-3 and answer the simulator-based questions.
3. Deploy the lab network in Task 4 and start the AttackBox; note the target subnet / connection_IP.
4. Run an ARP scan: sudo nmap -PR -sn <subnet>/24 and record how many hosts respond.
5. Run an ICMP echo scan: sudo nmap -PE -sn <subnet> and compare the result with the ARP scan.
6. Run ICMP timestamp (-PP) and ICMP Address-Mask (-PM) scans; note which ones are filtered.
7. Run TCP SYN (-PS), TCP ACK (-PA) and UDP (-PU) ping scans and compare their results and privilege requirements.
8. Optionally capture the scan traffic using tcpdump or wireshark on the tun0 interface.
9. Run nmap -sL <subnet>/24 for target listing and nmap -R <subnet>/24 for reverse-DNS lookup.
10. Complete the knowledge-check questions (Task 5-9) and

Room progress: 63%

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Task 1 Introduction

Task 2 Subnetworks

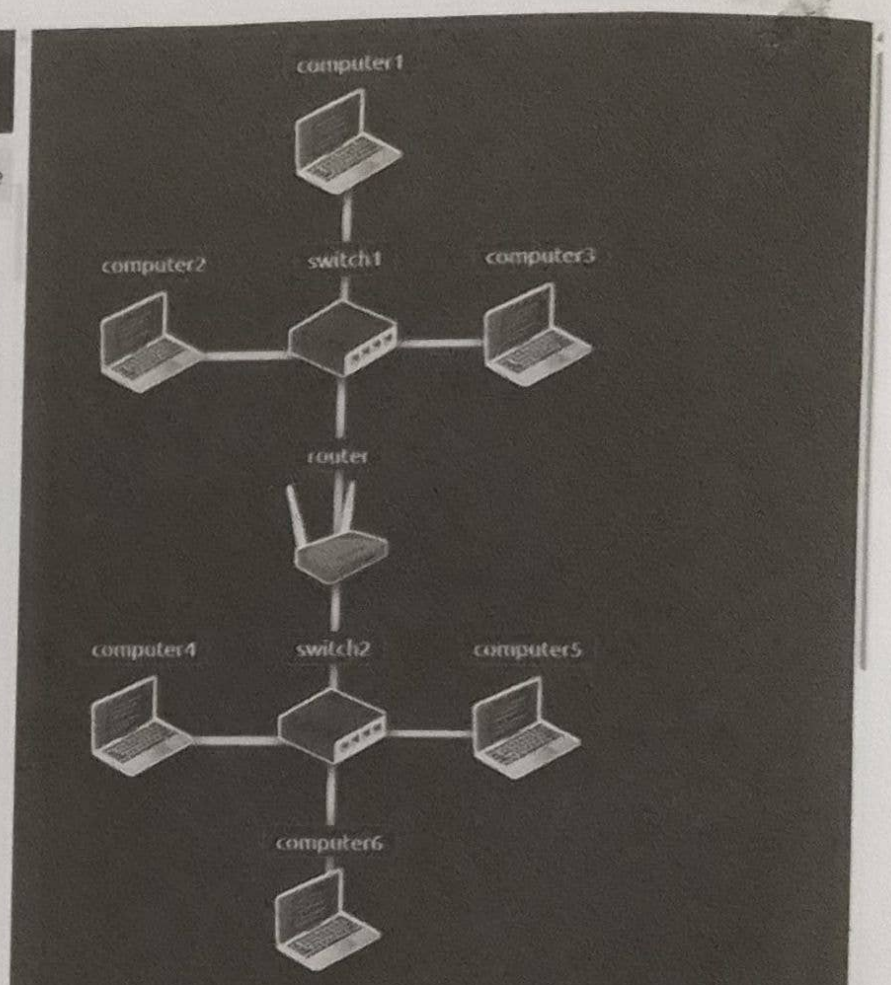
Task 3 Understanding Hosts Discovery through TCP/IP Layer

Task 4 Enumerating Targets

Task 5 Nmap Host Discovery Using ARP

Task 6 Nmap Host Discovery Using ICMP

We can ping every IP address on a target network and see who would respond to our `ping` (ICMP Type 8/Echo) requests with a ping reply (ICMP Type 0). Simple, isn't it? Although this would be the most straightforward approach, it is not always reliable. Many firewalls block ICMP echo; new versions of MS Windows are configured with a



Nmap: Network Simulator

Result:

Thus, Nmap host discovery scans were performed successfully using TCP SYN, TCP ACK, and UDP ping techniques, and the scan results and network traffic were observed.